

Week 12: Attention & Transformers

Deep Learning – IITM BS

Comprehensive Notes + Practice Problems

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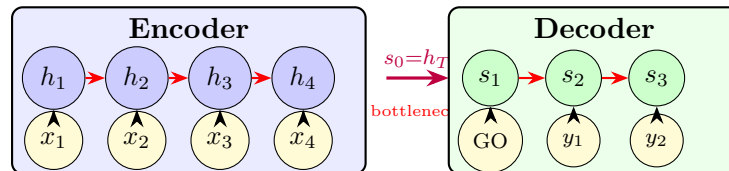
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1 Encoder-Decoder Framework

1.1 The Bottleneck Problem

In a basic encoder-decoder (seq2seq):

- Encoder reads entire input sequence $\rightarrow$ compresses into one vector h_T
- Decoder generates output conditioned on h_T
- **Problem:** All information squeezed into one fixed-size vector!
- Long sequences lose early information

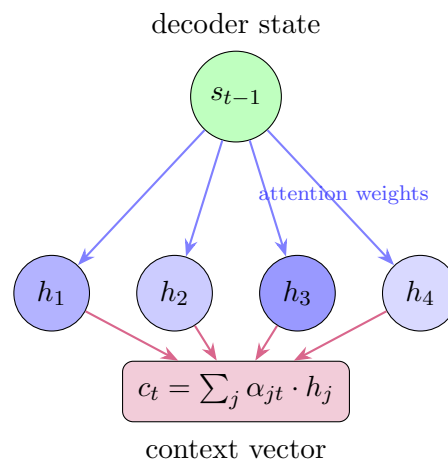


Solution: Instead of using only h_T , let the decoder **look at all encoder states** $h_1, h_2, \dots, h_T$. This is the **attention mechanism**.

2 Attention Mechanism

2.1 Core Idea

At each decoding step t , compute a **weighted sum** of all encoder hidden states, where weights reflect how “relevant” each encoder state is for the current decoding step.



Attention Computation – STEP BY STEP**Step 1:** Compute raw attention scores (energy):

$$e_{jt} = V_{\text{att}}^T \tanh(U_{\text{att}} \cdot s_{t-1} + W_{\text{att}} \cdot h_j)$$

Step 2: Normalize with softmax to get attention weights:

$$\alpha_{jt} = \frac{\exp(e_{jt})}{\sum_{j'=1}^T \exp(e_{j't})}$$

Step 3: Compute context vector (weighted sum of encoder states):

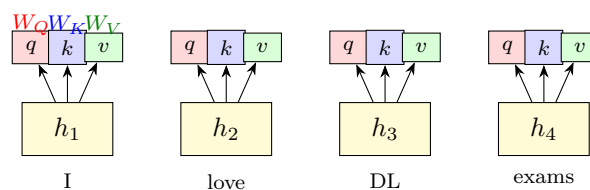
$$c_t = \sum_{j=1}^T \alpha_{jt} \cdot h_j$$

Step 4: Use context in decoder:

$$s_t = \text{RNN}(s_{t-1}, [e(\hat{y}_{t-1}), c_t])$$

Attention Score Worked Example
 $h_1 = [1, 0]^T, s_{t-1} = [1, 2]^T, U_{\text{att}} = W_{\text{att}} = I \text{ (identity)}, V_{\text{att}} = [1, 1]^T.$
 $U_{\text{att}} \cdot s_{t-1} = [1, 2]^T, W_{\text{att}} \cdot h_1 = [1, 0]^T$
 $\text{Sum} = [1 + 1, 2 + 0] = [2, 2]$
 $\tanh([2, 2]) = [0.964, 0.964]$
 $e_1 = V^T \tanh = [1, 1] \cdot [0.964, 0.964] = \mathbf{1.928}$
3 Self-Attention (Foundation of Transformers)**3.1 From RNN Attention to Self-Attention****RNN attention:** Decoder state s_{t-1} attends to encoder states h_j . States computed **sequentially**.**Self-attention:** Each position attends to **all other positions** in the *same sequence*. Computed in parallel!**3.2 Query, Key, Value**For each input embedding h_j , create three vectors:
 $q_j = W_Q h_j$ (query: “what am I looking for?”)

 $k_j = W_K h_j$ (key: “what do I contain?”)

 $v_j = W_V h_j$ (value: “what do I output?”)


3.3 Scaled Dot-Product Attention

Self-Attention Formula

Stack all queries, keys, values into matrices:

$$Q = W_Q H, \quad K = W_K H, \quad V = W_V H$$

where $H = [h_1, h_2, \dots, h_T]$ (columns are embeddings).

$$Z = \text{softmax}\left(\frac{Q^T K}{\sqrt{d_k}}\right) V^T$$

Why scale by $\sqrt{d_k}$? Without scaling, when d_k is large, dot products become very large $\rightarrow$ softmax saturates $\rightarrow$ vanishingly small gradients.

3.4 Attention Output Shape

Attention Output Dimensions

Given: $Q \in \mathbb{R}^{d_k \times t}$, $K \in \mathbb{R}^{d_k \times t}$, $V \in \mathbb{R}^{d_v \times t}$

$Q^T K \in \mathbb{R}^{t \times t}$ (attention score matrix)

$\text{softmax}(Q^T K / \sqrt{d_k}) \in \mathbb{R}^{t \times t}$

$V^T \in \mathbb{R}^{t \times d_v}$

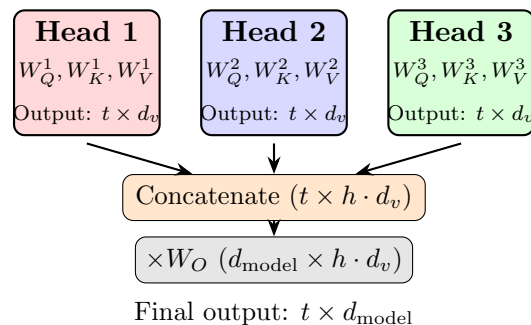
$$\text{Output} = \underbrace{(t \times t)}_{\text{attention}} \times \underbrace{(t \times d_v)}_{V^T} = t \times d_v$$

Total elements in output per head = $t \times d_v$

4 Multi-Head Attention (MHA)

4.1 Why Multiple Heads?

Same idea as multiple CNN filters: each head can learn **different types of relationships** (syntactic, semantic, positional, etc.).



MHA Parameter Count – EXAM FAVORITE

Given: d_{model} , h heads, $d_k = d_q = d_{\text{model}}/h$ (usually), d_v .

Per head: $W_Q \in \mathbb{R}^{d_k \times d_{\text{model}}}$, $W_K \in \mathbb{R}^{d_k \times d_{\text{model}}}$, $W_V \in \mathbb{R}^{d_v \times d_{\text{model}}}$

Per head params: $(d_k + d_k + d_v) \times d_{\text{model}} = (2d_k + d_v) \times d_{\text{model}}$

All h heads: $h \times (2d_k + d_v) \times d_{\text{model}}$

Output projection: $W_O \in \mathbb{R}^{d_{\text{model}} \times (h \cdot d_v)}$, params = $d_{\text{model}} \times h \cdot d_v$

$$\text{MHA params (no bias)} = h(2d_k + d_v) \cdot d_{\text{model}} + d_{\text{model}} \cdot h \cdot d_v$$

MHA Parameter Count Example

$t = 32$, $d_{\text{model}} = 64$, $h = 2$, $d_v = 16$.

$d_k = d_q = 64/2 = 32$.

Per head: $W_Q(32 \times 64) + W_K(32 \times 64) + W_V(16 \times 64) = 2048 + 2048 + 1024 = 5120$

$\times 2$ heads: $5120 \times 2 = 10240$

W_O : $64 \times (2 \times 16) = 64 \times 32 = 2048$

Total MHA = $10240 + 2048 = \mathbf{12288}$

5 Masked Self-Attention

Used in the **decoder** to prevent attending to future tokens during training.

Mask Matrix M ($t = 5$)

0					
0	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$
0	0	$-\infty$	$-\infty$	$-\infty$	$-\infty$
0	0	0	$-\infty$	$-\infty$	$-\infty$
0	0	0	0	$-\infty$	$-\infty$

Diagonal = 0 (can see self)

Upper = $-\infty$ (future blocked)

Lower = 0 (past visible)

Masked Attention Facts

- Sum of diagonal elements = **0** (each token can attend to itself)
- Masking: $\text{softmax}\left(\frac{Q^T K}{\sqrt{d_k}} + M\right) V^T$
- $e^{-\infty} = 0$, so masked positions get zero attention weight
- Total attention scores per head = $T \times T$
- Total across all: $B \times h \times T \times T$ (batch $\times$ heads $\times$ seq $\times$ seq)

6 Feed-Forward Network (FFN)

Applied **identically** and **independently** at each position after attention:

$$\text{FFN}(z) = \max(0, z \cdot W_1 + b_1) \cdot W_2 + b_2$$

FFN Parameters

$W_1 \in \mathbb{R}^{d_{\text{model}} \times d_{\text{ff}}}$, $b_1 \in \mathbb{R}^{d_{\text{ff}}}$, $W_2 \in \mathbb{R}^{d_{\text{ff}} \times d_{\text{model}}}$, $b_2 \in \mathbb{R}^{d_{\text{model}}}$

$$\text{FFN params} = 2 \cdot d_{\text{model}} \cdot d_{\text{ff}} + d_{\text{ff}} + d_{\text{model}}$$

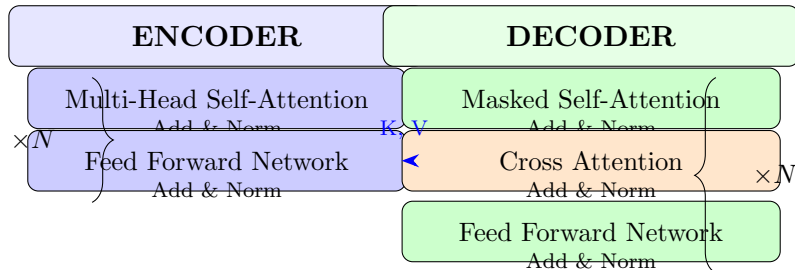
Typical: $d_{\text{ff}} = 4 \times d_{\text{model}}$ (e.g., $d_{\text{model}} = 512$, $d_{\text{ff}} = 2048$).

FFN Parameter Count Example

$d_{\text{model}} = 64$, $d_{\text{ff}} = 256$, with bias.

W_1 : $64 \times 256 = 16384$, b_1 : 256, W_2 : $256 \times 64 = 16384$, b_2 : 64.

Total = $16384 + 256 + 16384 + 64 = \mathbf{33088}$.

7 Full Transformer Architecture

- (P) Encoder Self-Attn → source attends to all source
 (Q) Masked Self-Attn → prevents future token access
 (R) Cross Attention → Q from decoder, K/V from encoder
 (S) FFN → position-wise after attention

Cross Attention in Decoder

In cross attention:

- **Keys** and **Values** come from the **encoder** output
- **Queries** come from the **decoder** (output of masked self-attention)
- This is how the decoder “reads” the encoder’s representation
- Each head has its own W_Q , W_K , W_V – they do NOT share!

8 Practice Problems

Problem 1: Attention Score [3 marks]

$$h_1 = [1, -1]^T, h_2 = [0, 2]^T, s_{t-1} = [1, 1]^T.$$

$$U_{\text{att}} = W_{\text{att}} = I \text{ (identity)}, V_{\text{att}} = [1, 1]^T.$$

Compute e_1 and e_2 , then compute the attention weights α_1 and α_2 .

Problem 2: Context Vector [2 marks]

Using the attention weights from Problem 1, compute the context vector $c_t = \alpha_1 h_1 + \alpha_2 h_2$.

Problem 3: Attention Output Shape [2 marks]

$t = 32$, $d_{\text{model}} = 64$, $h = 2$, $d_v = 16$. What is the total number of elements in the attention output for a **single** head?

Problem 4: MHA Parameters [3 marks]

$t = 32$, $d_{\text{model}} = 64$, $h = 2$, $d_v = 16$. Compute the total number of parameters in Multi-Head Attention (no bias).

Problem 5: FFN Parameters [2 marks]

$d_{\text{model}} = 64$, $d_{\text{ff}} = 256$, with bias. Compute total FFN parameters.

Problem 6: Masked Attention [2 marks]

Sequence length $T = 8$. What is the sum of all diagonal elements of the mask matrix? How many elements in the mask are $-\infty$?

Problem 7: Total Attention Scores [2 marks]

Batch size $B = 4$, sequence length $T = 12$, heads $h = 2$. How many total attention scores are computed?

Problem 8: Larger MHA [3 marks]

$d_{\text{model}} = 256$, $h = 8$, $d_v = d_k = 32$, no bias. Compute total MHA parameters.

Problem 9: Matching [3 marks, MCQ]

Match the attention mechanism with its description:

- (P) Multi-Head Attention (Encoder)
- (Q) Masked Multi-Head Self-Attention
- (R) Multi-Head Cross Attention
- (S) Feed Forward Network

- (1) Queries from decoder, K/V from encoder
- (2) Prevents access to future target tokens
- (3) Each source token attends to all source tokens
- (4) Applied position-wise after attention

- A. P→2, Q→3, R→1, S→4 B. P→3, Q→1, R→2, S→4
C. P→3, Q→2, R→1, S→4 D. P→4, Q→2, R→3, S→1

Problem 10: Conceptual MSQ [3 marks]

Which statements about multi-head attention are correct? (Select all.)

- A. Scaling by $\sqrt{d_k}$ prevents dot products from becoming too large.
- B. All heads share the same projection matrices for Q, K, V.
- C. If all heads learn identical projections, MHA behaves like single-head attention.
- D. Increasing heads always increases total parameters.

9 Solutions

Problem 1:

e_1 : $Us_{t-1} + Wh_1 = [1, 1] + [1, -1] = [2, 0]$. $\tanh([2, 0]) = [0.964, 0]$. $e_1 = [1, 1] \cdot [0.964, 0] = 0.964$.

e_2 : $Us_{t-1} + Wh_2 = [1, 1] + [0, 2] = [1, 3]$. $\tanh([1, 3]) = [0.762, 0.995]$. $e_2 = [1, 1] \cdot [0.762, 0.995] = 1.757$.

$\exp(0.964) = 2.624$, $\exp(1.757) = 5.795$. Sum = 8.419.

$\alpha_1 = 2.624/8.419 = \mathbf{0.312}$, $\alpha_2 = 5.795/8.419 = \mathbf{0.688}$.

Problem 2: $c_t = 0.312[1, -1]^T + 0.688[0, 2]^T = [0.312, -0.312]^T + [0, 1.376]^T = [\mathbf{0.312}, \mathbf{1.064}]^T$.

Problem 3: Output per head = $t \times d_v = 32 \times 16 = \mathbf{512}$ elements.

Problem 4: $d_k = 64/2 = 32$. Per head: $(32 + 32 + 16) \times 64 = 5120$. Two heads: 10240. W_O : $64 \times 32 = 2048$. Total = **12288**.

Problem 5: $2(64)(256) + 256 + 64 = 32768 + 256 + 64 = \mathbf{33088}$.

Problem 6: Diagonal sum = **0** (all zeros, token can see itself).

Number of $-\infty$ elements = upper triangle = $\frac{T(T-1)}{2} = \frac{8 \times 7}{2} = \mathbf{28}$.

Problem 7: $B \times h \times T \times T = 4 \times 2 \times 12 \times 12 = \mathbf{1152}$.

Problem 8: $d_k = 256/8 = 32$. Per head: $(32 + 32 + 32) \times 256 = 24576$. 8 heads: 196608. W_O : $256 \times 256 = 65536$. Total = **262144**.

Problem 9: C. $P \rightarrow 3$, $Q \rightarrow 2$, $R \rightarrow 1$, $S \rightarrow 4$.

Problem 10: A, C. B is false (each head has own projections). D is false (with fixed d_{model} , more heads means smaller $d_k = d_{\text{model}}/h$, so total Q/K/V params stay the same; only W_O is constant).